

From Microscope Photographs to Battery Physics: An Interpretable, Uncertainty-Aware Pipeline for Silicon–Graphite Anode Cross-Sections

micro2dfn pipeline — technical report

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Abstract

This report explains, end to end and in plain language, how we turned scanning electron microscope (SEM) photographs of a battery electrode into numbers that describe its internal structure, and then into physics-model estimates of what that structure might mean for battery behaviour. The dataset is 31 backscattered electron (BSE) images from three manufacturing batches of a silicon–graphite anode. The pipeline has five stages: (1) correct each photograph’s uneven lighting; (2) colour-code every pixel into one of three materials — pores, graphite, silicon — using one shared set of brightness cut-offs for all photos; (3) separate real silicon particles from bright imaging artefacts; (4) measure about thirty structural *markers* per image, each with an uncertainty band; (5) feed those markers into PyBaMM’s Doyle–Fuller–Newman (DFN) battery model, run repeatedly across the uncertainty ranges, to obtain consequence *indicators* such as swelling and stress during the cycle. Every step is deterministic and documented: no machine-learning black box is used anywhere in the main pipeline, and the measuring rule itself is frozen once calibrated on the approved reference batch, so arriving photos can never change how earlier batches were measured. The honest headline is two-fold. Compositionally, the three batches are indistinguishable — about 6% silicon-candidate material and about 10% resolved porosity in each. But we then discovered that the photographs were taken under at least a dozen distinguishable imaging conditions that line up with the batch labels almost as well as the material does. After controlling for that *acquisition* factor, no batch difference in the image set can currently be asserted — not because the batches are proven identical, but because the measurement cannot yet separate material from imaging session. The pipeline’s most defensible outputs are therefore: a measurement system that knows its own uncertainty, a documented confound that any future batch-comparison must control for, and a set of candidate signals — three unusually bright Batch 1 fields, a within-session-consistent InLens pore-contrast direction, and a model-conditional silicon stress difference — each labelled solid, possible, or can’t-tell rather than promoted to a verdict. All model outputs are reported as ranges and are explicitly labelled as indicators rather than predictions.

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1 Introduction — what problem are we solving?

1.1 A two-minute battery primer

A lithium-ion battery cell is a sandwich of two electrode coatings with a separator in between, all soaked in liquid electrolyte. The electrode we study here is the *negative electrode* (anode): a thin coating, roughly the thickness of a human hair, made of microscopic particles glued onto a metal foil.

The coating contains three things that matter to us:

- **Graphite particles** — the workhorse material. When the cell charges, lithium ions leave the other electrode, travel through the electrolyte, and slot themselves between graphite’s atomic layers.
- **Silicon particles** — a smaller fraction blended in because silicon can store roughly ten times more lithium per weight than graphite. The catch: silicon swells to about $3\times$ its size when it fills with lithium, which stresses everything around it.
- **Pores** — the empty space between particles. This is where the electrolyte lives. Pores are the “roads” that lithium ions travel along to reach the particles. Their size, shape and connectedness control how fast the cell can charge and how evenly it wears.

Because performance and lifetime depend on how these ingredients are arranged — not just how much of each there is — two coatings with identical recipes can behave very differently if they were mixed, coated or calendered differently. That is why manufacturers photograph polished cross-sections of electrodes with electron microscopes: the internal arrangement is directly visible.

1.2 What this study asks

We have BSE cross-section images from three manufacturing batches of a silicon–graphite anode (7, 7 and 17 photographs). We ask:

1. Do the batches differ structurally, and if so, in which measurable features and by how much?
2. How confident can we be in those differences?
3. What could the structural differences physically mean, if we feed the measured structure into a standard battery physics model?

Just as importantly, we require the answer to be *explainable*: every number in this report can be traced back to a formula applied to a specific image, and every claimed difference comes with an uncertainty band.

1.3 What “interpretable” means here

Much image analysis today hands the problem to a neural network that outputs a score with no explanation. We deliberately did the opposite. Every stage of the pipeline is a named, classical method whose behaviour can be inspected:

raw image → lighting correction (Gaussian flat-field)
→ pixel classification (shared Otsu thresholds)
→ particle splitting (watershed)
→ object classifier (interior brightness + shape)
→ geometric/statistical markers (counting, lengths, distances)
→ batch statistics (medians, bootstrap intervals)
→ physics model (published DFN equations, PyBaMM)

If a reader disagrees with any step, they can change exactly that step — a threshold, a filter size, a parameter — and re-run one command to see how every downstream number responds.

2 The data

Table 1: Dataset summary. All images are 8-bit greyscale TIFFs at ≈ 25 nm per pixel, each frame $\sim 7000 \times 1800$ –2300 pixels ($\approx 175 \times 50$ μm of material).

Batch	BSE images	Notes
Batch_1	7	contains two low-contrast photographs (flagged automatically)
Batch_2	7	–
Batch_3	17	–
Total	31	

Each image is a *backscattered electron* (BSE) micrograph: brightness corresponds loosely to average atomic number, so the three ingredients appear as three brightness levels:

- **Dark pixels** = pores (empty space).
- **Grey pixels** = graphite plus the carbon-binder matrix — these cannot be told apart in BSE and are treated as one “bulk” phase throughout.
- **Bright pixels** = silicon-containing material.

Two framing limitations, stated up front. First, the images do not show the metal foil or the coating’s outer surface, so we cannot measure total electrode thickness from them, and we do not know with certainty which edge faces the separator — this is why directional markers are used as *ratios*, never as absolute directions. Second, at 25 nm/pixel we reliably resolve pores and particles down to about 0.35 μm ; finer structure exists but is invisible to us. We call the measured pore fraction “resolved porosity” and never equate it with true porosity.

3 Method — every step, explained

3.1 Step 1 — evening out the lighting

SEM images often have a slow brightness gradient across the frame (the middle is brighter than the corners). If we thresholded the raw image, a grey graphite region in a bright corner could look like silicon.

The fix is called *flat-field correction*: we blur a downsampled copy of the image with a wide Gaussian filter, which captures only the slow lighting gradient and none of the real particles, then divide the raw image by that background. Afterwards, “the same material” has “the same brightness” in every corner of the image — which is the precondition for honest thresholding.

3.2 Step 2 — colouring in the photograph (segmentation)

We assign every pixel to pore, bulk or silicon using *multi-Otsu thresholding*: an algorithm that looks at the brightness histogram and finds the two cut-off values that best separate the three brightness peaks.

The single most important design choice: the thresholds and the object classifier are fitted *once, on the 17 baseline (Batch_3) images only*, saved to `recipe.json`, and reused verbatim on every later run ($t_{\text{pore}} = 0.711$, $t_{\text{Si}} = 1.335$, $t_{\text{core}} = 1.550$). If each photo were allowed its own thresholds, a genuinely silicon-rich batch would simply re-calibrate itself into looking normal — like letting a suspect grade their own exam. Fitting on the reference batch — rather than all discovered images — means a future batch is measured against today’s ruler, not a slightly moved one; `run_manifest.json` records the recipe hash, code hashes and every input-image hash so each report identifies exactly which run produced it. This freezes the *software* ruler. Section 4.4 documents the harder problem it cannot solve alone: the *imaging* ruler — session noise, contrast and detector settings — also varied between photographs, and no software freeze can subtract a hardware drift after the fact.

After thresholding we remove specks below the resolution floor — silicon objects under 150 px and pores under 20 px (roughly $0.35\text{ }\mu\text{m}$) are absorbed into their surroundings — so that single-pixel noise can never become a “particle”.

3.3 Step 3 — splitting touching particles (watershed)

Silicon particles often touch in the image and get merged into one blob, which would corrupt every particle-size statistic. We use the standard *watershed* algorithm: the image is treated as a landscape where distance from the nearest non-silicon pixel is altitude; the landscape is “flooded” and ridges form where touching particles meet, splitting them at the natural necks. This yields 4313 separate bright objects in total.

3.4 Step 4 — real silicon versus bright junk

Not every bright object is a silicon particle. In low-contrast photographs, stretched binder regions, particle edges and fines can pass the brightness threshold but are not real particles. We therefore run an *object-level classifier*: for each bright object we measure its interior brightness (ignoring its rim) and its compactness, and compare the interior brightness against a cut-off $t_{\text{core}} = 1.550$ derived from the 5th-percentile interior brightness of large, obviously-real particles. Objects that fail are labelled `bright_fine` rather than `si_particle`.

The split: **4313 silicon particles vs. 1714 bright-fine objects**. Both populations are kept and reported separately (Figure 1), so the reader can see exactly what the classifier did rather than trust it. This matters: in the two flagged low-contrast photos the “apparent silicon” drops by roughly half after classification (raw bright fraction $\approx 10\%$ $\rightarrow$ classified silicon $\approx 5\%$) — i.e. half the bright material there was artefact. Had we not split it, those photos would have silently doubled the measured silicon.

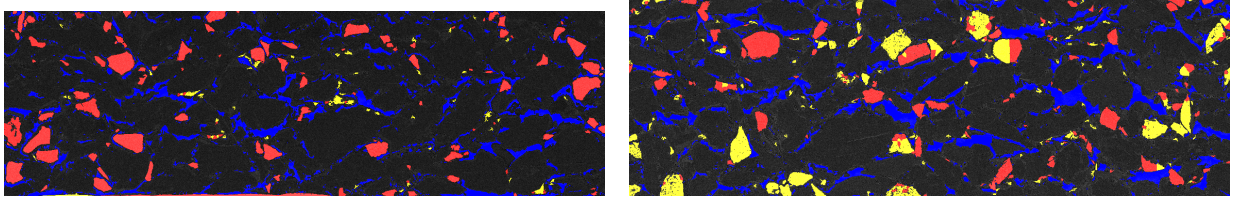


Figure 1: Classification overlays. **Left:** a normal-contrast image (Batch_2). **Right:** one of the two flagged low-contrast Batch_1 photos. Colour code: **red** = classified silicon particles, **yellow** = bright-fine material (bright but failed the interior-brightness test), **blue** = pores, dark grey = graphite/binder bulk. Note how much more yellow the problem photo contains — exactly the artefact the classifier exists to catch.

3.5 Step 5 — measuring the structure (the markers)

From the three-colour map we compute roughly thirty *markers* per image — single numbers that summarise one structural aspect. A reader with no materials background only needs the intuition behind each group; Appendix A defines every marker individually.

3.5.1 Group A — composition: “how much of each ingredient?”

The simplest possible measurement: count pixels of each colour. A classical result called the *Delesse principle* (1847) guarantees that, for a random cut, the area fraction of a phase equals its volume fraction — so counting pixels on a slice legitimately estimates 3-D composition. We report `pore_frac_resolved` (the pores we can see), `si_frac_total`, `gr_frac`, and `bright_fine_frac` (the artefact budget — itself a data-quality guard).

3.5.2 Group B — silicon geometry: “how big are the particles?”

For every classified silicon particle we measure its area and shape, giving the diameter distribution (`si_d10/d50/d90`), the large-particle tail (`si_frac_gt3um`, `_gt5um`), solidity and aspect ratio. The 3-D particle radius is corrected for the fact that a random slice rarely cuts a sphere through its middle — on average a 2-D section reads $\sim 21\%$ small (the Wicksell/Saltykov correction, standard textbook stereology). Surface area per volume (`si_sv`) comes from the Crofton formula, $S_V = \frac{4}{\pi} L_A$: total boundary length on the slice, rescaled. Why it matters: silicon’s surface is where the degrading SEI layer forms, and particle size sets both the distance lithium must diffuse inside the particle and the stress it feels on swelling.

3.5.3 Group C — graphite: “how is the matrix arranged?”

Same integral-geometric measures applied to the bulk–pore interface: `gr_sv` and an effective particle radius $R_{\text{eff}} = 3V/S$ (`gr_radius_eff_um`). Graphite flakes are not spheres, so this is honestly labelled an effective (diffusion-equivalent) radius. Mean horizontal versus vertical chord lengths (`gr_chord_ratio_xz`) reveal whether flakes lie flat — as they typically do after calendaring.

3.5.4 Group D — pores and transport: “how hard is the road for lithium?”

Pore size (`pore_d50`), count density, and *anisotropy* (`pore_anisotropy` > 1 means pores lie flat, forcing ions onto longer through-plane detours). Elongated near-vertical pores are counted separately as `crack_frac_v` — mechanical tears are also express ion lanes.

Tortuosity (how much longer a real path through the pores is than a straight line) is attacked two ways. First, a geodesic proxy: the shortest path through the pore maze versus a straight

line, which feeds our *Bruggeman-equivalent exponent* `bruggeman_b_eff` (through $\tau = \varepsilon^{1-b}$). Second, a true diffusion-solve tortuosity (finite-difference Laplace solve through the pore phase, the same approach as `TauFactor/porespy`). That second solver *honestly reports not-measurable* on every image: the resolved pore network never forms a connected path across any 2-D section (`pore_spans_* = 0`). This is expected, not a bug — through-plane connections leave the image plane, and much of the real transport network lives in sub-resolution pores. We keep the spanning check (`percolation_reach_2d`, `pore_largest_region_frac`) as relative markers rather than inventing finite numbers.

Finally, silicon accessibility: mean distance from each Si particle to the nearest pore (`si_pore_dist_mean_um`), the share of Si fully enclosed in bulk (`si_enclosed_share`), and per-particle pore coverage of each Si boundary (`si_lowcoverage_share`). Silicon sealed away from electrolyte-filled pores is effectively dead material.

3.5.5 Group E — swelling and heterogeneity: “where will it crack?”

Silicon swells $\sim 280\%$ on lithiation. `swelling_budget` asks, locally, whether the nearby pore space can absorb each particle’s expansion; `swelling_deficit_area_frac` is the fraction of the image where demand exceeds supply — a spatial risk map, not just an average. `si_clustering_R` (Clark–Evans nearest-neighbour ratio, edge corrected; < 1 = clumped, ≈ 1 = random, > 1 = evenly spread) and `si_patchiness` catch mixing problems that averages hide.

3.5.6 Group F — imaging guards

Per-photo quality markers (`si_bulk_contrast`, noise, sharpness, shading, detector presence, `is_problem_photo`) ensure an “imaging” difference cannot masquerade as a “material” difference.

3.5.7 Group G — assumed, never measured

Electrode thickness (75 μm), the sub-resolution porosity correction (+0.15–0.30), the Bruggeman default (1.5) and all material constants (diffusivities, densities, kinetics — from the published `Chen2020_composite` and `OKane2022` parameter sets). These are declared assumptions and are swept over, never baked in silently.

3.6 Step 6 — uncertainty: bands, not points

Every marker is measured on every image, so a batch gives a small sample (7, 7 or 17 values). We report the batch *median* with its interquartile range, and compare batches with a *bootstrap* 95% confidence interval on the median difference: resample both batches’ image lists thousands of times, recompute the difference each time, and take the central 95% of the resulting distribution. A difference is called *clear* only when that interval excludes zero. This uses no assumption about the data being normally distributed — appropriate for small, possibly skewed samples.

3.7 Step 7 — from images to a battery model (DFN)

The Doyle–Fuller–Newman model is the standard physics model for lithium-ion cells: coupled equations describing how lithium moves through electrolyte and particles, how it reacts at surfaces, and how the cell voltage responds. PyBaMM is a widely used open-source implementation. Crucially, DFN is *homogenised*: it does not want a 3-D picture; it wants a handful of summary numbers — which is exactly what our markers are. The mapping:

Table 2: How measured markers become PyBaMM inputs. “Assumed” entries are declared and swept, not measured.

Marker	PyBaMM parameter
porosity_est (resolved + correction)	Negative electrode porosity
si_frac_of_solid $\times$ si_accessible_frac remainder	Secondary: ... active material volume fraction Primary: ... active material volume fraction
si_radius_3d_um	Secondary: ... particle radius [m]
gr_radius_eff_um	Primary: ... particle radius [m]
bruggeman_b_eff	Negative electrode Bruggeman coefficient (electrolyte)
assumed 75 μm	Negative electrode thickness [m]

Because every input carries uncertainty, we do not run the model once. We run a 9-point sweep per batch spanning the marker bands — 27 solves in total — and report each output as a band, not a point. The model used is a composite silicon/graphite DFN with particle swelling and cracking, SEI growth, and stress-driven loss of active material enabled; three 1C charge–discharge cycles per solve. Convergence was 27/27.

4 Results

4.1 Photo-quality check

Two Batch_1 photographs (img_4ih2ggld, img_5n1q8atc) were automatically flagged as low-contrast. As Figure 1 shows, their raw bright fraction ($\approx 10\%$) halves once bright-fine artefacts are excluded. They are kept in the analysis (flagged, not deleted) because the classifier removes exactly the artefact class that made them suspect.

4.2 Marker distributions

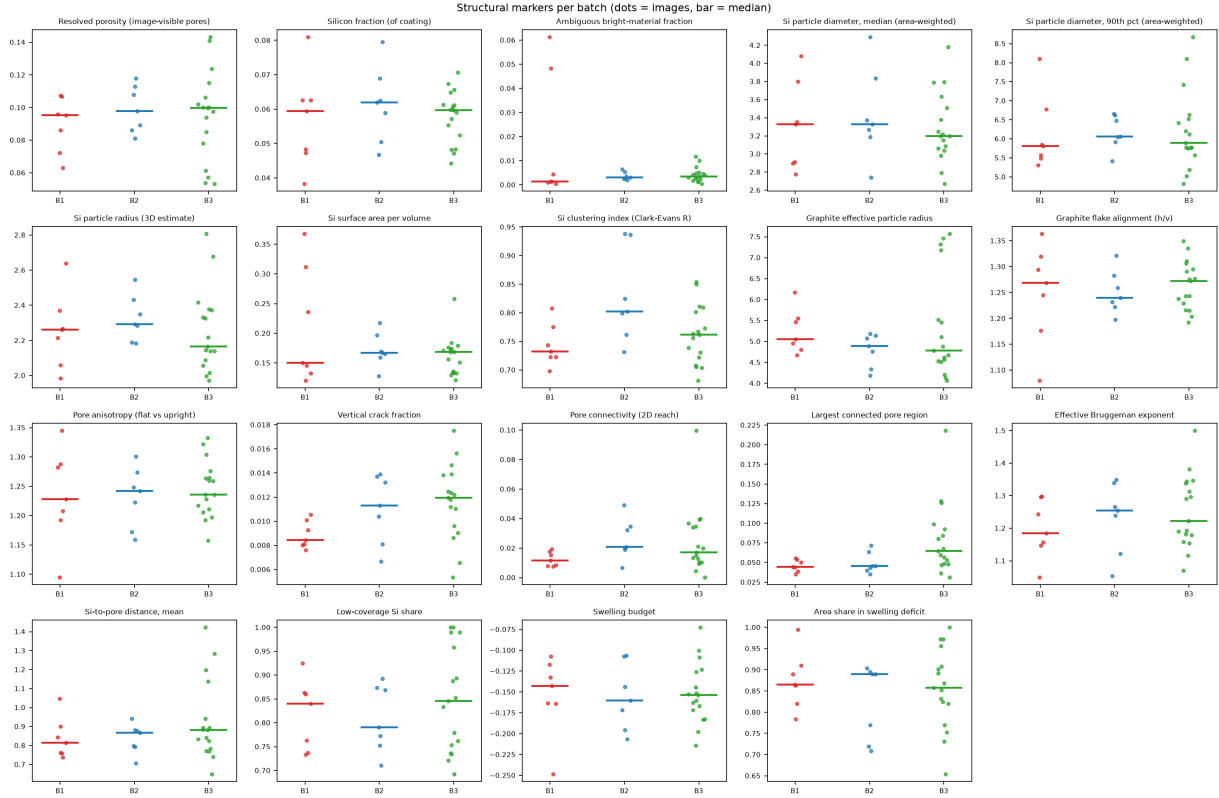


Figure 2: Every key marker, per batch: each dot is one image, the bar is the batch median (B1 red, B2 blue, B3 green). Non-measurable markers (the diffusion-solve tortuosity, infinite on all images) are omitted rather than drawn as empty panels.

How to read Figure 2. Each panel is one marker; each dot is one photograph; the short bar is the batch’s middle value. Two things to look for: (1) *do the clouds overlap?* — in nearly every panel the three colours intermingle, which is why composition markers show no clear differences; (2) *is the bar offset even where clouds overlap?* — e.g. in “Si clustering index” the red cloud sits slightly below the blue one consistently, which is the kind of small but systematic shift the bootstrap flags as a *candidate* — whether it survives the acquisition control is the subject of Section 4.4. Note also Batch_3’s wider spread in several panels — with 17 photos it shows the most internal variation, which is normal for a larger sample rather than a defect of the batch.

Table 3: Batch medians for key markers (full table and per-image values in `markers_*.csv`; definitions in Appendix A).

Marker	Batch_1	Batch_2	Batch_3	Unit
Resolved porosity	0.095	0.098	0.100	–
Silicon fraction	0.059	0.062	0.060	–
Bright-fine (artefact)	0.001	0.003	0.003	–
Si D50 (area-weighted)	3.33	3.33	3.20	μm
Si D90	5.81	6.06	5.90	μm
Si 3-D radius (est.)	2.26	2.29	2.17	μm
Si surface/volume	0.150	0.167	0.169	μm^{-1}
Si clustering index R	0.73	0.80	0.76	–
Graphite eff. radius	5.05	4.89	4.78	μm
Pore anisotropy	1.23	1.24	1.24	–
Vertical crack fraction	0.0085	0.011	0.012	–
2-D pore reach	0.012	0.021	0.017	–
Largest pore region	0.045	0.046	0.065	–
Bruggeman-equiv. b	1.18	1.25	1.22	–
Si-to-pore distance	0.82	0.87	0.88	μm
Si accessible fraction	0.61	0.55	0.57	–
Low-coverage Si share	0.84	0.79	0.85	–
Swelling budget	–0.143	–0.160	–0.154	–
Contrast (guard)	2.24	2.45	2.39	–

Reading Table 3 as a non-specialist: the three batches are *compositionally near-identical* — resolved porosity 9.5–10.0%, silicon $\approx 6\%$. The differences that do exist are *spatial*: how the silicon is arranged and how the pore network connects.

4.3 The four candidate differences — and the test they still have to pass

Out of every marker $\times$ every batch pair, exactly four comparisons survive the bootstrap test — the 95% confidence interval on the median difference excludes zero. That number is itself informative: with ~ 20 markers and 3 batch pairs, a handful of surviving differences means the batches are *mostly* alike. We originally reported these four as “the differences”. They are better described as *candidates*, because they were computed ignoring how the photographs were taken — and Section 4.4 shows why that omission matters. We keep them here, fully analysed, because they are what any acquisition-controlled re-test would need to re-establish.

Table 4: Clear batch differences (bootstrap 95% CI on median difference). The sign gives the direction of (first – second). Plain-English effect sizes are discussed below.

Pair	Marker	Difference	95% CI
B1 – B2	Si clustering index R	–0.070	[–0.213, –0.009]
B1 – B2	Pore connectivity (2-D reach)	–0.009	[–0.026, –0.002]
B1 – B3	Vertical crack fraction	–0.0035	[–0.0054, –0.0011]
B1 – B3	Largest connected pore region	–0.020	[–0.048, –0.001]

All four point at **Batch_1**. Two say its silicon sits differently; two say its pore network is wired differently. Below, each difference is expanded: what the marker is, how big the effect really is, what could cause it in manufacturing, and why it matters.

4.3.1 Difference 1 — silicon is more clumped in Batch_1 (Clark–Evans R : 0.73 vs. 0.80)

What it measures. Pick every silicon particle’s centre, and measure how far each one is from its nearest neighbour. If particles were sprinkled completely at random, the average nearest-neighbour distance takes a known value; R is the ratio of observed to expected. $R = 1$ is random; $R > 1$ means particles repel each other into an even grid; $R < 1$ means they clump. R has a second intuitive reading: at $R = 0.73$, a typical particle’s nearest neighbour is $\sim 27\%$ closer than random chance predicts.

How big is the effect. Batch_1’s median is $R = 0.73$ — every one of its 7 photos sits in the range 0.70–0.81, i.e. *consistently* clumped. Batch_2’s median is 0.80 (range 0.73–0.94) — some photos nearly even. The medians differ by 9%, and the bootstrap interval barely clears zero, so the direction is reliable but the size is modest: we are talking about visibly lumpier mixing, not separated islands. Crucially the *amount* of silicon is identical — this is purely about where it landed.

Physical meaning. Clumped silicon means clumped swelling. When charging, each Si particle inflates to $\sim 3\times$ its volume; a patch of particles expanding together concentrates stress like several people pushing on the same spot of a wall — local cracking and delamination become more likely than the average silicon content would suggest.

Plausible manufacturing cause. Mixing/dispersion differences — silicon aggregates that were not fully broken up, or slurry rheology that let particles re-aggregate before drying. It is a *process* signature, not a recipe signature. Figure 3 makes it visible.

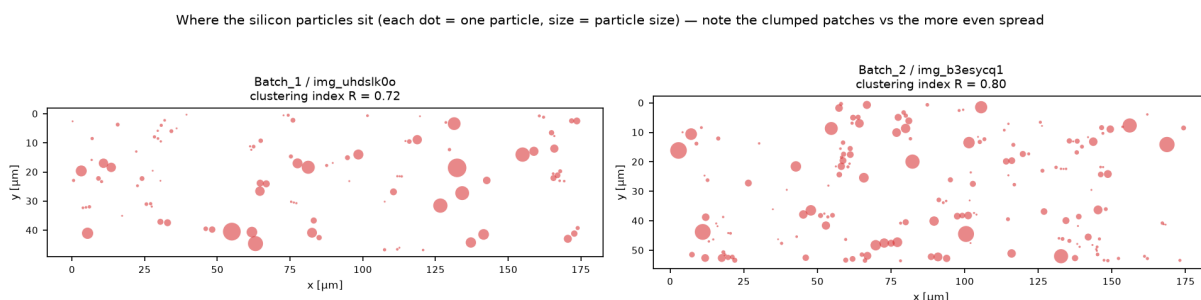


Figure 3: Silicon particle positions in the most “typical” clean photo of Batch_1 (left) and Batch_2 (right); each disc is one particle, sized by its measured diameter. Left shows the batch-median clustering index $R = 0.72$: particles gather into visible patches with gaps between. Right ($R = 0.80$) is more uniformly sprinkled. Same silicon *amount*, different silicon *arrangement*.

4.3.2 Difference 2 — Batch_1’s pore network reaches less far (2-D percolation reach: 0.012 vs. 0.021)

What it measures. Take every pore pixel that belongs to a connected network spanning the image edge-to-edge, and ask what fraction of all pore pixels that is. It is a coarse “how wired-together is the pore maze” score. (The stricter test — a single unbroken path across the whole section — fails on *all* images; see the tortuosity note in Section 3.5.4.)

How big is the effect. Batch_2’s median reach is about $1.8\times$ Batch_1’s (0.021 vs. 0.012). In plain terms, in a typical Batch_2 photo roughly twice as much of the visible pore space belongs to a long-range connected channel as in Batch_1. The batches overlap at the tails, which is why the CI is wide, but the typical values are nearly a factor of two apart.

Physical meaning. Pores are the electrolyte’s roads. A better-connected network means ions can reach deep into the coating through fewer dead ends — faster charging capability and more

uniform utilisation. Batch_1's sparser resolved network implies more of its interior is served only by sub-resolution pores we cannot see.

Plausible cause / caveat. Compaction (calendering) pressure, particle packing, or simply which parts of the coating happened to be sectioned. Honest caveat: this is a 2-D measurement of a 3-D network — real connectivity is certainly better than any section shows, and the comparison is only meaningful *between* batches measured identically.

4.3.3 Difference 3 — Batch_3 has more vertical crack-like pores (crack fraction: 0.012 vs. 0.0085)

What it measures. The area share of pores that are both elongated and oriented steeply — the signature of mechanical tears running through the coating's thickness rather than along it.

How big is the effect. Batch_3's median is about **40% higher** than Batch_1's (0.0119 vs. 0.0085) — small in absolute terms ($\sim 1\%$ of image area) but consistent enough across images to clear the test.

Physical meaning — genuinely ambiguous, and we say so. Vertical cracks cut both ways. As *damage*, they are pre-existing failure initiation sites — bad. As *ion lanes*, they are express routes that short-circuit the tortuous pore maze — arguably good for rate capability. The data cannot choose between these readings; we report the difference and its two-sided meaning rather than a verdict.

Plausible cause. Calendering damage, drying stress, or particle fracture during pressing — all of which are process history, not composition.

4.3.4 Difference 4 — Batch_1's largest pore region is smaller (largest-region share: 0.045 vs. 0.065)

What it measures. The single biggest connected pore blob, as a fraction of all pore area. If the pore network were one big web, this would approach 1; a fragmented network leaves it near zero.

How big is the effect. Batch_3's typical largest pore region is about **45% bigger** than Batch_1's. Same direction as difference 2, measured a different way — which is reassuring: two independent connectivity markers agree that Batch_1's resolved pore network is the most fragmented. When two markers tell the same story, it is less likely to be noise or a quirk of one particular definition.

Physical meaning. Again: electrolyte access. A fragmented network has more isolated pockets — regions ions can only reach through connections too fine to see here.

4.3.5 Reading the four candidates together — unblocked

Individually each candidate is modest. Together they sketch a coherent story: **Batch_1's structure looks spatially lumpier** — silicon gathered into patches, pores broken into more isolated pieces — while Batches 2 and 3 look smoother and better connected, with Batch_3 carrying slightly more (mechanical-history) vertical cracking. Every candidate is a *spatial arrangement* pattern; none is a *composition* difference. That pattern would point at process variables — mixing, coating, calendering — *if* it survives the imaging control test in the next section.

4.4 The acquisition confound — when the camera is part of the measurement

What we found while auditing our own conclusions. The 31 photographs were not all taken under the same conditions. Grouping them by frame size, pixel size, detector set, noise level and contrast reveals 13 distinct *acquisition groups* — inferred imaging sessions, since real session metadata does not yet exist. The groups predict every acquisition property almost perfectly (0–7% residual variance for noise, sharpness, bulk brightness and contrast), and they cut across the batch labels: the approved reference, Batch_3, appears in seven different groups, and five groups contain photographs from more than one batch.

Why this matters — the measurement changes with the session. Table 5 partitions each feature’s variance into “how much is left over after removing group means” versus “batch means”. For *count*-type markers the group nearly explains everything: within one group, photographs from *different* batches have almost identical pore counts (4.3% residual), while the same batch spread across sessions swings by ± 20 –50%. For *fraction*-type markers the opposite holds: batch labels explain almost nothing, and real specimen-to-specimen spread shows through even inside one imaging session (23% residual).

Table 5: Variance partition: which variable organises each measurement better — imaging group or batch? “Residual after group” is the variance left once group means are removed; lower means the group explains more. Count-type markers read the session; fraction-type markers read the specimen.

Marker	Residual after group	Residual after batch	Reads mostly
Pore count / Mpx	4.3%	52.7%	the session
Noise level	0.0%	50.8%	the session
Image sharpness	0.1%	48.4%	the session
Bulk brightness	2.9%	80.1%	the session
Si–bulk contrast	7.4%	71.0%	the session
Pore area fraction	23.2%	95.2%	the specimen
Silicon area fraction	20.9%	89.0%	the specimen

Why counts are fragile and fractions are not. A count depends on where object boundaries land, and boundary position shifts with acquisition contrast and sharpness: a 16-pixel pore that blurs into two 8-pixel pores doubles its contribution to the count while the total pore *area* is conserved. That is why a minimum-size cut-off did not rescue the count — the session effect is not tiny specks at the detection floor, it is edge drift across the whole size range. Area fractions are merge-invariant by construction. The practical consequence for any future analysis: *fractions are primary markers; counts are within-session diagnostics only*.

The blocking test — what survives. We re-ran the batch comparisons using only photographs taken inside the five groups that mix batches (a stratified permutation that shuffles batch labels only within groups, after centring each marker on its group median). The available evidence is thin — Batch_1 and Batch_3 share exactly one group, with one photo each — but the result is unambiguous in direction:

Table 6: Within-session batch comparisons (group-centred values in mixed groups; stratified permutation p -values). Every unblocked candidate collapses; in the single session shared by Batch_1 and Batch_3, Batch_1 actually has *fewer* pores than Batch_3 (123 vs. 141 per Mpx).

Marker	B1–B3 (p)	B1–B2 (p)	B2–B3 (p)
Pore count / Mpx	−4.1 (0.37)	+0.3 (0.96)	−4.4 (0.36)
Raw bright fraction	−0.006 (0.39)	−0.009 (0.18)	+0.004 (0.45)
Total Si fraction	−0.008 (0.16)	−0.004 (0.59)	−0.004 (0.37)
Resolved pore fraction	−0.011 (0.13)	−0.010 (0.29)	−0.001 (0.86)

Does the pipeline see material or sessions? Two more checks make the picture sharper. (a) A classifier given only acquisition information — frame height, noise, contrast — predicts the batch label with 68% leave-one-out accuracy, *above* the 55% majority baseline; the full image-feature set manages only 42%. The imaging conditions carry more apparent batch information than the material features do. (b) The three new-image assignments each landed in the one batch that shares the photo’s acquisition group — consistent with either a correct batch match or pure session matching; the blind protocol was correct, but the outcome cannot distinguish the two.

What this does and does not prove. It does not prove the batches are identical — the within-session test is severely under-powered (the key Batch_1–Batch_3 pair shares a session exactly once). It proves that the current dataset cannot separate material differences from acquisition conditions, and therefore that no batch-comparison claim from count-type or unblocked markers is presently supportable. The definitive fix is in the measurement system: acquisition metadata from the imaging lab (dates, voltage, working distance, detector settings, specimen IDs) to convert the 13 inferred groups into confirmed sessions, or ideally a re-imaging run with batches interleaved within each session.

4.5 DFN consequence indicators

Table 7: DFN model indicators under the repaired model — medians and [min, max] over the 17-point uncertainty sweep per batch (51/51 solves converged, primary **accessible** variant). Indicators are read at *in-cycle peaks*, not after the final rest; all three batches use one shared cathode sized on the reference at each sweep index. These compare batches under identical model assumptions; they are *indicators*, not *predictions*.

Indicator	Batch_1	Batch_2	Batch_3
Cycle-5 discharge capacity [Ah]	5.59	5.68	5.48
Peak cell thickness change [μm]	12.3	12.2	12.1
Peak Si surface stress [MPa]	10.0	9.53	10.8
Min. neg. surface potential [mV]	−8.2	−8.2	−8.6

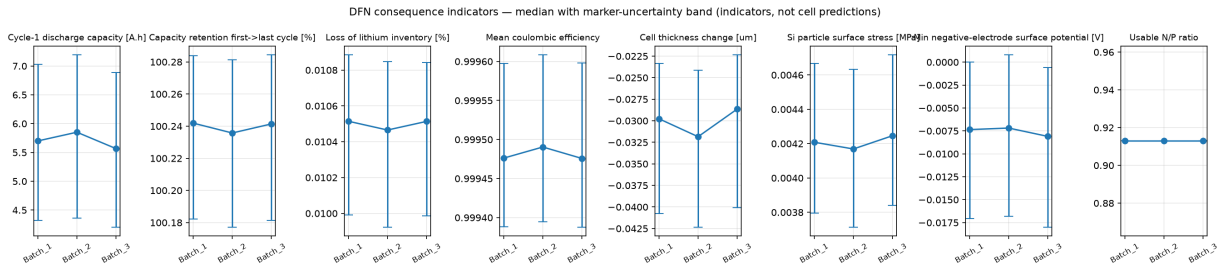


Figure 4: Consequence indicators per batch: point = median across the sweep, bar = min–max band produced purely by marker uncertainty.

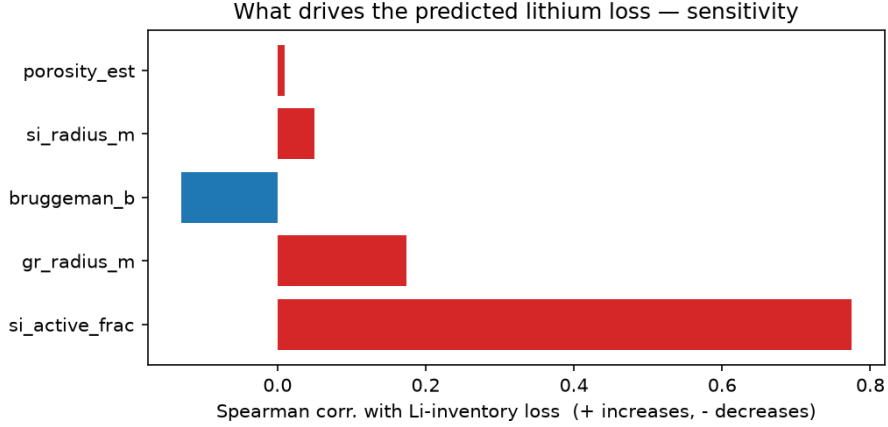


Figure 5: Sensitivity analysis (Spearman correlation of each swept input with peak silicon stress). Shows which measurement uncertainty dominates the model answer — the modelled reachable-silicon share dominates ($\rho \approx -0.93$), then assumed porosity and transport; silicon particle size contributes nothing ($\rho \approx 0$).

4.5.1 How to read Table 7 and Figure 4

Each number is a median over 17 model runs spanning the marker uncertainty band, and every batch is evaluated with the *same* cathode at each sweep index — so batch differences are read as paired differences at fixed assumptions, not as median gaps. The rows, in plain terms:

- **Cycle-5 discharge capacity (~5.5–5.7 Ah).** Predicted output after five 1C cycles. Paired differences (+0.11 Ah for B1–B3, +0.15 Ah for B2–B3, –0.05 Ah for B1–B2) all sit below the 0.25 Ah threshold declared before the runs — and the swept assumptions alone move the answer by ~1.14 Ah (IQR). Below threshold, honestly reported.
- **Peak cell thickness change (~12 μm during discharge).** The *maximum* swelling reached mid-cycle, not the relaxed end-state — an earlier version of this analysis read the wrong moment and saw nothing. Paired differences ($\leq 0.15 \mu\text{m}$) stay below the declared 0.5 μm threshold.
- **Peak silicon surface stress (9.5–10.8 MPa during discharge).** The one comparison that *exceeds* its declared threshold (0.5 MPa) on all three pairs: B2–B3 = –1.60 MPa (–15% of the B3 value), B1–B3 = –0.95 MPa (–9%), B1–B2 = +0.53 MPa (+5%), with paired confidence bands excluding zero. Inside the model’s physics this ordering is driven almost entirely by the *modelled* share of electrically reachable silicon (Spearman $\rho \approx -0.93$ with `si_accessible_frac`); silicon particle size contributes nothing ($\rho \approx 0$). Two caveats are mandatory: the carrier input is a 2-D contact proxy, not measured connectivity, and the comparison inherits the acquisition confound — every input marker came from the same session-entangled images.
- **Minimum surface potential (–8 to –9 mV; plating context).** Every run of every batch dips slightly below 0 V anode surface potential during charge. Under the corrected rule — a warning counts only where the reference cell stays above zero under the same protocol — this is attributed to the shared assumed geometry (75 μm thickness, generic kinetics), not to any batch. Kept as a model-diagnostic, excluded from the batch verdict.
- **Lithium loss, retention, Coulombic efficiency — diagnostics only.** Nothing ages measurably in five cycles (retention $\approx 100\%$, LLI $\approx 0.01\%$, CE ≈ 0.9995 in all batches), so these are reported but explicitly *not compared*. An earlier pipeline version compared them anyway — and multiplied the lithium loss by a factor of ~3600 through a units bug. Both errors are recorded rather than hidden.

What the model adds — and what it cannot do. The DFN is a consequence model: it translates measured inputs into physical responses and adds no measurement of its own. Its one defensible contribution is the stress ordering above — a hypothesis, in plain words, that “Batch_2’s silicon may sit in a slightly more electrically accessible configuration than Batch_3’s, which would spread lithiation over more particles and lower per-particle stress”. The input markers that carry it are each individually below significance; the model integrates several weak, correlated shifts through a steep response. That makes it a signal worth testing directly in the image layer — not a measured fact, and not a batch verdict.

5 What the results mean, physically

5.1 Five insights, in plain terms

1. **Composition is indistinguishable — the only batch-comparative claim that survives.** Silicon-candidate fraction (5.95 / 6.14 / 6.03%), resolved pore fraction ($\sim 10\%$), particle size and surface area all overlap. The honest phrasing is “no detectable difference”: with seven photographs per batch, gaps under roughly two area-points would be invisible. The earlier “excess silicon” alarm was a threshold disagreement between two measurement recipes, not material — and the three-label ledger keeps it that way.
2. **Three Batch_1 fields carry unusually high bright-material area — real as an image observation, unresolved as material.** Three of seven Batch_1 photos hold 10.7–12.5% segmented bright area against the Batch_3 median of 7.7%. Two of the three are the low-contrast fields where much of that material is classified *uncertain bright* rather than candidate silicon — and in the one session where a flagged field shares a group with a Batch_2 photo, the two are nearly identical (10.7% vs. 10.3%). Solid as an observation of what is in the images; “can’t tell” as a batch claim until acquisition and chemistry are resolved.
3. **The camera is part of the measurement — the study’s own central finding.** Section 4.4 is not a caveat, it is a result: measured pore *counts* organise almost perfectly by imaging group (4% residual) and barely by batch (53%), while area *fractions* carry the specimen signal. The robust-measurement lesson — report fractions, demote counts — is a methodological result any comparable pipeline can reuse. Every unblocked batch candidate, including the four in Section 4.3, collapses under within-session blocking; the test is under-powered (the key pair shares one session), so the correct label is “can’t tell”, not “identical”.
4. **Every batch sits in a quantified swelling deficit — a dataset-level finding immune to the confound.** Silicon swells $\sim 2.8\times$ on lithiation, and in most 12.5 μm windows the resolved pore space cannot absorb that expansion; only ~ 1.5 –2.1% of candidate-particle boundary lies within 75 nm of a resolved pore. For a dense calendered coating this is normal — but it means the silicon’s electrolyte supply runs through sub-resolution porosity the images cannot see. Stated as a measured blind spot, not a defect: boundary contact, wetting and electrochemical activity are different quantities. Figure 6 shows the deficit tiles tracking the silicon.
5. **The widest error bar is still upstream of everything.** The sensitivity analysis ranks assumed true porosity far above every measured input (assumptions move predicted capacity by ~ 1.14 Ah against ≤ 0.15 Ah of paired batch difference), and the label-uncertainty bracket — classified silicon versus all uncertain-bright counted — alone moves the stress answer by up to 1.8 MPa, comparable to the batch stress difference itself. The two highest-value next

measurements are acquisition metadata (to break the session–batch tangle) and a measured porosity; both beat more photos.

5.2 The exploratory detector layer — what InLens and ETD added

Every photograph also carries two surface-sensitive detector channels (InLens and ETD), perfectly co-registered with the BSE frame (0-pixel shift on all 34 fields). We deliberately keep them *out* of the frozen segmentation — surface topography is a different physical quantity than atomic-number contrast, and their acquisition settings differ by batch — but as an exploratory layer they answered three questions BSE alone could not:

- **Pore contrast (label: possible).** Batch_1’s pores read darkest in InLens in all three sessions it shares with another batch — the only candidate signal whose direction survived blocking. With $n = 3$ matched observations and InLens gain itself differing by batch (median brightness ≈ 89 for Batch_3 vs. ~ 110 for Batches 1–2), this is a follow-up signal for controlled re-imaging, not a measured pore depth.
- **The “uncertain-bright” label is doing real work.** The material the classifier refuses to call silicon is at-or-above classified-silicon brightness in both surface channels, with about twice the surface texture — thin, edge-rich bright material rather than bulk grains. That validates keeping it as its own label; it does not prove it is silicon (edge and topography also read bright in surface channels), so the upper bracket remains a bracket.
- **A strong vertical texture exists in every image** (autocorrelation lengths $\sim 3\text{--}5\times$ longer along the vertical axis, plus Fourier agreement) and InLens does not exaggerate it — arguing against it being a pure surface-relief artefact. Its origin (calendering texture, or something else) remains open.

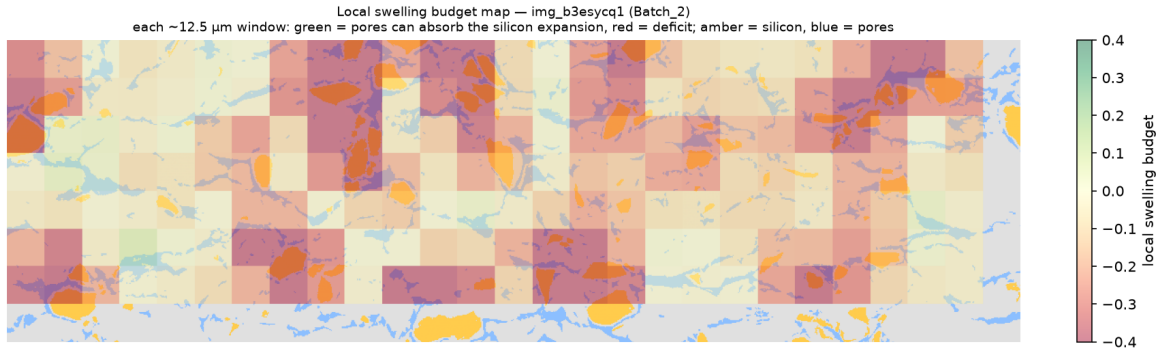


Figure 6: The swelling budget made spatial: each coloured tile is a $\sim 12.5\mu\text{m}$ window scored on “visible pore space minus silicon expansion demand”. Green windows have slack; red windows are in deficit. The red tiles visibly track the amber silicon particles — exactly the mechanism the marker encodes: stress concentrates *where the silicon is*, not uniformly. Underneath: amber = silicon, blue = pores.

5.3 What the model can and cannot say about the batches

Worth stating plainly: on three of four indicators the batches are indistinguishable, and on the fourth — peak silicon stress — the model reports a real, threshold-exceeding ordering that is nonetheless *conditional* twice over: on the model’s physics (which weights reachable-silicon share heavily, $\rho \approx -0.93$), and on image inputs that are themselves acquisition-entangled. The honest reading is a hypothesis to take back to the image layer — “Batch_2’s silicon may be in a more accessible configuration” — not a verdict. The bands, meanwhile, tell the durable truth: assumption and label uncertainty move every indicator more than the batches do.

5.4 What this analysis did detect, and what it would take to go further

The pipeline caught: a compositional identity result with explicit detection bounds; three unusually bright Batch_1 fields (real as an observation, unresolved as material); a systematic imaging artefact that would have doubled apparent silicon in two photos; a universal swelling-deficit regime; a count-versus-fraction measurement lesson; and — most importantly — the acquisition confound itself, quantified rather than suspected. What it could not detect: any batch difference separable from imaging session. The highest-leverage next steps, in order: (1) acquisition metadata or an interleaved re-imaging run — converts the confound from “can’t tell” into a controlled variable; (2) a measured electrode porosity and thickness (removes the widest model band); (3) EDS on a reviewed subset (converts “bright” into a chemical name); (4) electrochemical ground truth, a few cells per batch, to calibrate the indicators against reality.

6 Limitations — what we cannot claim

- **Acquisition and material are not adequately separated — the dominant limitation.** The inferred imaging groups are unevenly spread across batches, and the key Batch_1–Batch_3 pair shares a session exactly once. No batch-difference claim is currently supportable from count-type or unblocked markers; “can’t tell” is the correct status, not “identical”. The groups themselves are inferred (frame size, pixel size, noise, contrast) and need session metadata for confirmation.
- **“Bright” is not “silicon”.** BSE contrast identifies atomic-number-bright material; it cannot distinguish silicon from silicon-oxide-rich phases, edge artefacts or other heavy phases. EDS on a reviewed subset is the gate, and every silicon-fraction number in this report carries that asterisk.
- **Absolute values are illustrative; batch differences are the defensible signal.** We resolve only $\approx 10\%$ porosity against a true 25–40%; thickness is assumed; material constants are generic literature values. All three are swept, but they still cap absolute accuracy.
- **No 3-D connectivity.** The resolved pore network does not span any 2-D section; true through-plane tortuosity would need tomography or a reconstruction branch (SliceGAN + TauFactor), treated as an optional sensitivity analysis, not ground truth.
- **BSE cannot see** binder distribution, the SEI layer, lithium inventory, electrolyte wetting, or foil corrosion.
- **Model outputs are indicators under stated assumptions,** not validated predictions of capacity, cycle life or failure. They inherit the acquisition confound through their inputs, and there is no electrochemical ground truth in this dataset.
- **Statistical power** is limited by image counts (7/7/17) and, more sharply, by the near-total lack of shared imaging sessions between batches — the binding constraint on every within-session test.

7 Reproducibility

The frozen recipe is fitted once on the baseline batch and reused verbatim; every run writes a manifest identifying the exact ruler, code and inputs that produced it:

```
.venv/bin/python run_dfn.py --baseline Batch_3 --out dfn_output # fit+freeze
```

```
.venv/bin/python run_dfn.py --recipe recipe.json --reuse      # cached rerun
.venv/bin/python analyze_channels.py                        # InLens/ETD layer
.venv/bin/python analyze_spatial.py                       # spatial markers
```

dfn_output/ contains: `report.md` (readable summary), `markers_*.csv` (per image), `objects_*.csv` (per detected object), `batch_summary.csv`, `dfn_paired_differences.csv` (every pair-indicator verdict), `recipe.json` (frozen cut-offs), `run_manifest.json` (recipe/code/image hashes), and `dfn_params_*.json` (exact simulator inputs per run). The acquisition audit produces `image_acquisition.csv` (one row per photograph: frame size, pixel size, percentiles, noise, brightness, contrast, group ID) plus the variance-partition and blocking tables quoted in Section 4.4; the detector-layer audit produces `channel_features.csv` and `channel_report.md`.

A Marker glossary

Marker	Plain meaning
<code>pore_frac_resolved</code>	Fraction of the image that is visible empty space. Real porosity is higher — small pores are below resolution.
<code>porosity_est</code>	Resolved porosity plus a declared literature-based correction (+0.15–0.30); the value actually sent to the model, always as a range.
<code>si_frac_total</code> <code>gr_frac</code>	/ Area fractions of silicon / graphite-bulk in the coating.
<code>bright_fine_frac</code>	Share of bright material that failed the silicon test — an artefact budget and image-quality guard.
<code>si_d10/d50/d90_um</code>	10th/50th/90th-percentile silicon particle diameters (equivalent circles, area-weighted).
<code>si_radius_3d_um</code>	Estimated 3-D sphere radius, corrected for random slicing ($\approx 1/0.785\times$). Model input.
<code>si_sv_um-1</code>	Silicon surface area per volume, from boundary length (Crofton). Sets SEI-forming area.
<code>si_solidity/aspect</code>	How compact and how elongated silicon particles are — honesty terms for the spherical-particle model assumption.
<code>gr_radius_eff_um</code>	Diffusion-equivalent graphite “particle” radius, $3V/S$. Model input; flakes are not truly spherical.
<code>gr_chord_ratio_xz</code>	Horizontal vs. vertical chord lengths through bulk — flake alignment.
<code>pore_d50_um</code> , <code>pore_count_density</code> <code>pore_anisotropy</code>	Median pore size and pores per area. > 1 means pores lie flat — longer ion detours through the coating.
<code>crack_frac_v</code>	Area share of elongated near-vertical pores — tear/express-lane proxy.
<code>percolation_reach_2d</code> , <code>pore_largest_region_frac</code>	How much of the pore network belongs to big connected regions (2-D proxy only).
<code>tau_fdm_*</code> , <code>pore_spans_*</code>	Diffusion-solve tortuosity and spanning check — honestly <i>not measurable</i> here (no spanning path).
<code>bruggeman_b_eff</code>	Bruggeman exponent implied by the geodesic tortuosity proxy; model input.
<code>si_pore_dist_mean_um</code>	Mean distance from silicon to nearest pore — silicon far from pores is poorly wetted.

Marker	Plain meaning
si_accessible_frac, si_enclosed_share, si_lowcoverage_share	Share of silicon with/without electrolyte access — “dead silicon” proxies; accessible fraction scales the model’s active material.
corr_len*_um	Characteristic domain sizes of each phase (two-point correlation decay).
swelling_budget, swelling_deficit_area_frac	Local pore buffer minus Si expansion demand; deficit = image fraction where demand wins.
si_clustering_R, si_patchiness	Spatial statistics: $R < 1$ clustered, ≈ 1 random; patchiness = sub-region variance.
si_bulk_contrast, noise_mad, sharpness_*, is_problem_photo	Photo-quality guards — protect markers from imaging artefacts.